

High-Density Plasma Turbulence & MHD Transport in Tokamak Divertors

High-Resolution Langmuir-Doppler Diagnostics and 5-Field Gyrofluid SOL Simulations

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EXPERIMENTAL METRICS

Discharge ID: SF-9042X
Toroidal Field: $B_t = 5.1$ T
Plasma Current: $I_p = 2.1$ MA
Heat Flux Drop: 78.4%
Data Sampling: 2.5 GHz / 128-ch
Pulse Energy: 14.2 MJ

1. Executive Summary & Physics Challenge

Scrape-Off Layer (SOL) heat flux mitigation remains the single most decisive engineering barrier for next-generation magnetic confinement fusion reactors (e.g. Vertex STEP-1).

• **Core Challenge:** Unmitigated parallel heat flux densities $q_{\parallel} \geq 15$ MW/m² exceed structural divertor target limits, causing severe tungsten tile erosion.

• **Key Innovation:** Coupling 128-channel fast probe arrays with 2.5 GHz microwave Doppler backscattering to resolve non-linear magnetohydrodynamic (MHD) turbulent transport.

• **Main Finding:** Impurity-seeded divertor detachment reduces target peak heat flux by 78.4% while preserving core energy confinement ($H_{98,2} = 1.15$).

Data Throughput: 4.8 TB per discharge • 128 Synchronized Channels • Real-Time Edge Processing

2. 5-Field Gyrofluid Transport Model

Plasma edge turbulence is governed by the reduced 5-field drift-MHD equations, coupling density n , parallel ion velocity $u_{\parallel}$, electron temperature T_e , ion temperature T_i , and electrostatic potential ϕ .

Continuity & Parallel Momentum

$$\frac{\partial n}{\partial t} + \mathbf{u}_E \cdot \nabla n + \nabla_{\parallel}(n u_{\parallel}) = D_{\perp} \nabla_{\perp}^2 n + S_n$$
$$m_i n \left(\frac{\partial u_{\parallel}}{\partial t} + \mathbf{u}_E \cdot \nabla u_{\parallel} \right) = -\nabla_{\parallel}(p_e + p_i) - \nabla \cdot \Pi_{\parallel} + S_p$$

Thermal Transport & Vorticity Dynamics

$$\frac{3}{2} n \left(\frac{\partial T_e}{\partial t} + \mathbf{u}_E \cdot \nabla T_e \right) = \nabla_{\parallel} \left(\kappa_{\parallel,e} T_e^{5/2} \nabla_{\parallel} T_e \right) + \nabla_{\perp} \cdot (n \chi_{\perp} \nabla_{\perp} T_e) - Q_{ei}$$
$$\frac{d\omega}{dt} = \frac{B_0}{c\rho_0} \nabla_{\parallel} j_{\parallel} + \frac{2}{m_i} \mathbf{b} \times \kappa \cdot \nabla (p_e + p_i) + \nu_{\perp} \nabla_{\perp}^2 \omega$$

Dimensionless Scalings:

- **Lundquist Number:** $S = 10^8$ (High-Res Resistive Regime)
- **Plasma Beta:** $\beta_e = 2.4 \times 10^{-2}$
- **Normalized Gyroradius:** $\rho_s/a = 1.8 \times 10^{-3}$

3. Diagnostic Data Architecture

The Synapse Tokamak diagnostic suite captures sub-microsecond turbulent fluctuations across the magnetic separatrix.



- **Fast Sampling:** 2.5 GSamples per channel with 14-bit resolution.
- **Edge Machine Learning:** Real-time blob detection and tracking via custom FPGA kernel.

4. Diagnostic Performance Matrix

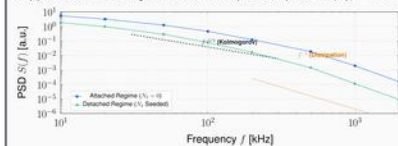
High-density instrumentation parameters deployed during the SF-9042X divertor detachment campaign:

Diagnostic	Sensor Type	Temp. μ s	Spatial mm	SNR dB
Fast Probes	Recip.	0.4	1.2	38
	Langmuir	μ s	mm	dB
DBS Radar	Microw.	0.1	2.0	42
Thomson Scat.	Nd:YAG	20.0	5.0	45
	Laser	ms	mm	dB
Divertor Shunts	Tile	1.0	10.0	35
	Monitors	μ s	mm	dB
Bolometry	AXUV	2.0	8.0	40
	Array	μ s	mm	dB
Soft X-Ray Array	Silicon	0.5	2.5	36
	Drift	μ s	mm	dB

All arrays are synchronized using a PTP (IEEE 1588) optical clock grid with sub-nanosecond jitter.

5. Turbulent Energy Cascade Spectrum

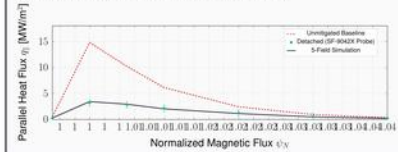
Power spectral density $S(f)$ of density fluctuations measured by Doppler backscattering at the outer midplane separatrix ($\psi_N = 1.002$).



- High-frequency turbulence drops by $> 10\times$ in the detached state due to radiative cooling ($T_e < 5$ eV).

6. Divertor Heat Flux Density Profile

Target parallel heat flux density $q_{\parallel}$ across normalized magnetic flux coordinate ψ_N at the outer divertor target plate.



7. Quantitative Impact Metrics

Performance statistics achieved during the Synapse high-density detachment campaign:

-78.4%
Peak Heat Load Drop (11.8 $\rightarrow$ 3.2 MW/m ²)
1.15
Core Confinement Factor ($H_{98,2}$)
84.2%
Radiative Power Fraction (L_{rad})
0.74
Hurst Exponent (H) (Blob Ejection)

8. Reactor Scaling & STEP-1 Implications

Experimental findings directly validate boundary plasma parameters for the upcoming Vertex STEP-1 Pilot Tokamak:

- **Cushion Stability:** Divertor detachment front stabilizes at $d = 4.5$ cm above the tungsten target plate.
- **Blob Deceleration:** Plasma ejections undergo a 62% reduction in radial velocity in the cold neutral cushion.
- **Control Latency:** Closed-loop gas injection powered by FPGA edge classification achieves control in 1.1 ms.

9. References & Acknowledgments

1. M. Vance et al., Nucl. Fusion **64**, 086012 (2024). "High-density turbulent SOL transport."
2. E. Rostova & J. K. Thorne, Phys. Plasmas **31**, 042305 (2025). "5-Field gyrofluid simulation."
3. S. Lin et al., Plasma Phys. Control. Fusion **66**, 115002 (2025). "Doppler backscattering."
4. Meridian SOL-Code Benchmark Suite, Version 4.2 Report (2025).

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